We have,

$$P(A_1) = 0.40, \quad P(B/A_1) = 0.04$$

$$P(A_2) = 0.60, \quad P(B/A_2) = 0.05$$

$$P(A_2/B) = \frac{P(A_2)\,P(B/A_2)}{P(A_1)\,P(B/A_1) + P(A_2)\,P(B/A_2)}$$

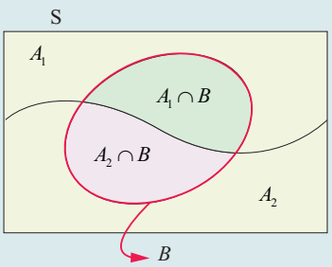
$$P(A_2/B) = \frac{(0.60)(0.05)}{(0.40)(0.04)+(0.60)(0.05)} = \frac{15}{23}.$$

Example 12.27

A construction company employs 2 executive engineers. Engineer-1 does the work for 60% of jobs of the company. Engineer-2 does the work for 40% of jobs of the company. It is known from the past experience that the probability of an error when engineer-1 does the work is 0.03, whereas the probability of an error in the work of engineer-2 is 0.04. Suppose a serious error occurs in the work, which engineer would you guess did the work?

Solution

Let A_1 and A_2 be the events of job done by engineer-1 and engineer-2 of the company respectively. Let B be the event that the error occurs in the work.



We have to find the conditional probability

$P(A_1/B)$ and $P(A_2/B)$ to compare their errors in their work.

From the given information, we have

$P(A_1) = 0.60, \quad P(B/A_1) = 0.03$

$P(A_2) = 0.40, \quad P(B/A_2) = 0.04$

A_1 and A_2 are mutually exclusive and exhaustive events.

Applying Bayes' theorem,

$$P(A_1/B) = \frac{P(A_1)\,P(B/A_1)}{P(A_1)\,P(B/A_1) + P(A_2)\,P(B/A_2)}$$

$$= \frac{(0.60)(0.03)}{(0.60)(0.03)+(0.40)(0.04)}$$

$$P(A_1/B) = \frac{9}{17}.$$

$$P(A_2/B) = \frac{P(A_2)\,P(B/A_2)}{P(A_1)\,P(B/A_1) + P(A_2)\,P(B/A_2)}$$

$$P(A_2/B) = \frac{(0.40)(0.04)}{(0.60)(0.03)+(0.40)(0.04)}$$

$$P(A_2/B) = \frac{8}{17}.$$

Since $P(A_1/B) > P(A_2/B)$, the chance of error done by engineer-1 is greater than the chance of error done by engineer-2. Therefore one may guess that the serious error would have been be done by engineer-1.

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